



AI/ML PROJECT REPORT

Automated Chest X-ray Disease Classification

Abstract

This project presents an automated deep learning system for multi-class chest X-ray disease classification using EfficientNet-B0. The pipeline includes preprocessing, data augmentation, class imbalance handling, model training, classical model comparisons, and explainability using Grad-CAM. Medical evaluation metrics such as AUC, precision, recall, and F1-score show strong performance for major lung conditions, supported with interpretable visualizations.

Report Outline

- Introduction and motivation.
- Dataset description and preprocessing.
- Exploratory Data Analysis.
- Data augmentation pipeline.
- EfficientNet-B0 architecture and training methodology.
- Classical ML comparison.
- Explainability using Grad-CAM.
- Streamlit web application interface.
- Final results, conclusion, and future work.

1. Introduction

Chest X-rays remain the most frequently used imaging tool for diagnosing lung diseases. This project builds an automated, explainable diagnostic system using EfficientNet-B0 to detect pathologies such as pneumonia, cardiomegaly, emphysema, nodules, and more.

Objectives:

- Develop a deep learning model using EfficientNet-B0.
- Handle dataset imbalance using weighted loss.
- Evaluate using AUC, precision, recall, and F1-score.
- Provide explainability using Grad-CAM.
- Deploy a user-friendly Streamlit web application.

2. Dataset Description and Preprocessing

The dataset contains 112,120 chest X-rays across 14 diseases + No Finding.

Preprocessing Steps:

- Resize images to 224×224.
- Convert grayscale → RGB.
- Apply ImageNet normalization.
- Train/Validation/Test split (patient-wise).

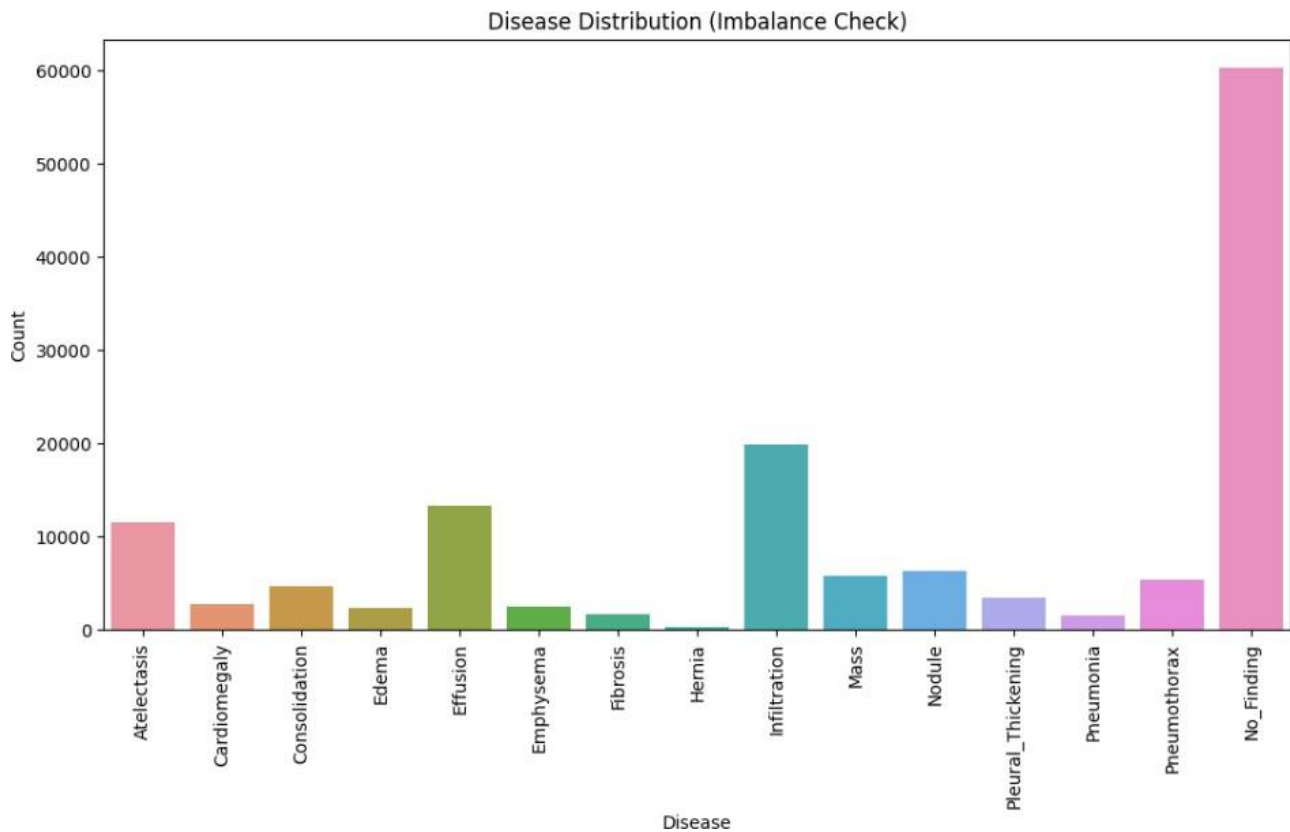


Figure 1: Disease distribution showing strong class imbalance across categories.

3. Exploratory Data Analysis (EDA)

EDA helps understand demographic patterns and frequency distribution.

Key Observations:

- No Finding dominates the dataset heavily.
- Hernia, Fibrosis, and Pneumonia have fewer samples.
- Age of patients varies significantly across diseases.

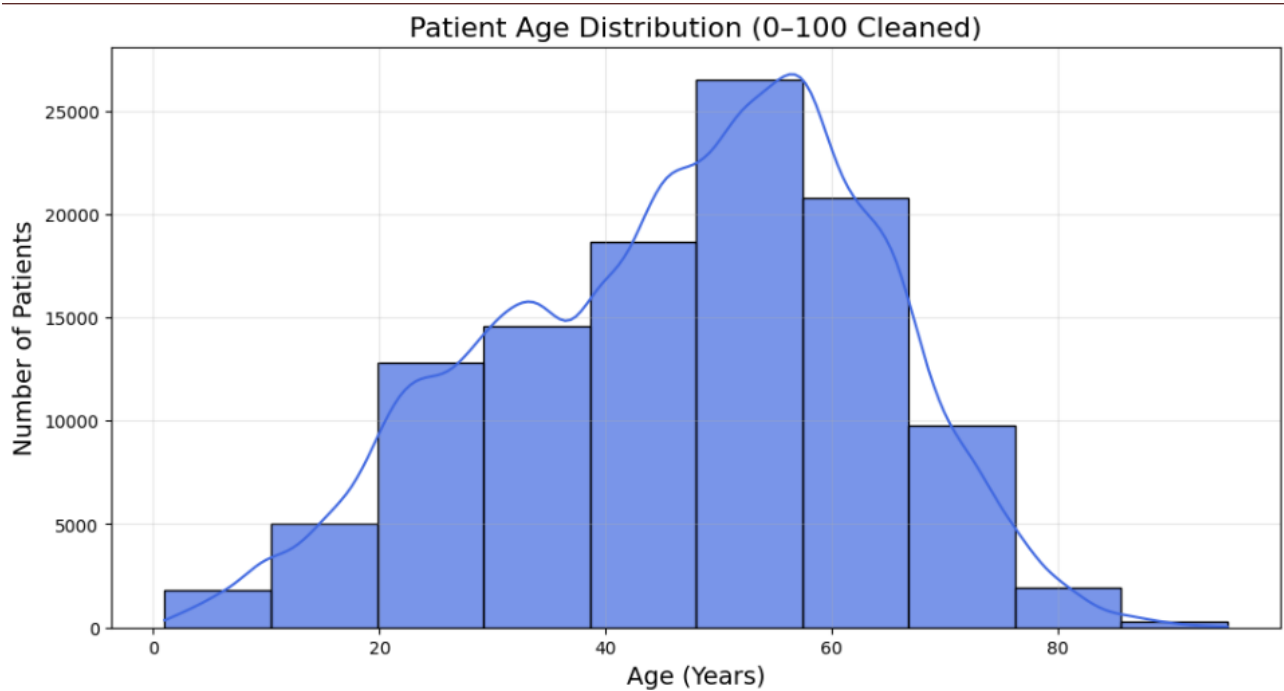


Figure 2: Age distribution patterns across chest diseases.

4. Data Augmentation

Augmentation ensures the model generalizes well despite imbalance.

Techniques Used:

- Random rotation ($\pm 10^\circ$)
- Horizontal flip
- Brightness/contrast jitter
- Scaling and shifting

5. EfficientNet-B0 Architecture

EfficientNet-B0 provides high accuracy with fewer parameters.

Custom Classifier Head:

- Global Average Pooling
- Dropout (0.4)
- Fully connected layer

Training Configuration:

- Loss: Weighted Cross Entropy
- Optimizer: Adam
- Scheduler: Cosine Annealing

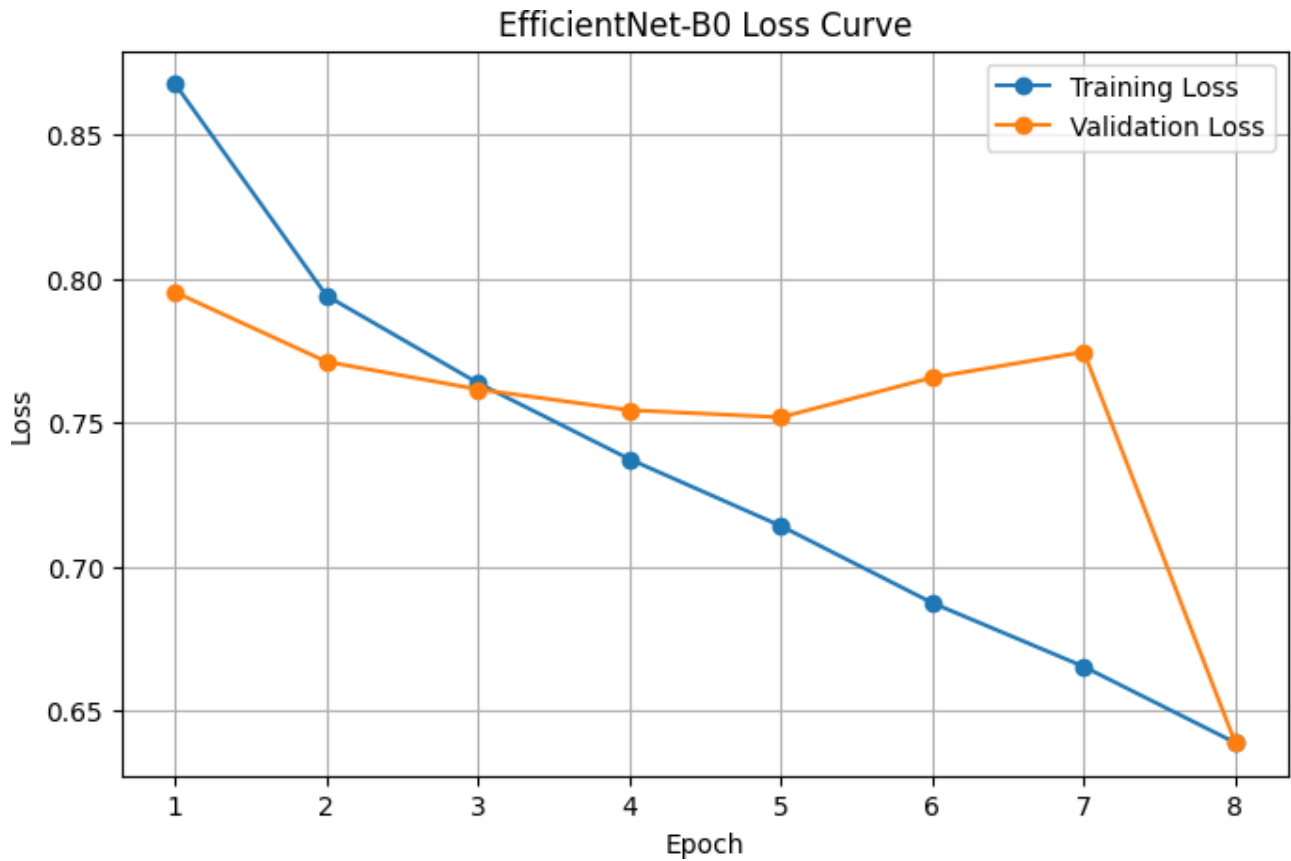


Figure 3: Training and validation loss curve.

6. Classical Model Comparison

```
==== FINAL AUC SCORES ====
Logistic Regression (Torch)   : 0.4730
Random Forest (NumPy)        : 0.6946
CatBoost                      : 0.6730
Naive Bayes                   : 0.5848
MLP (Torch)                   : 0.5169
```

Figure 4: AUC comparison of classical Machine Learning models.

7. Final Per-Class Metrics

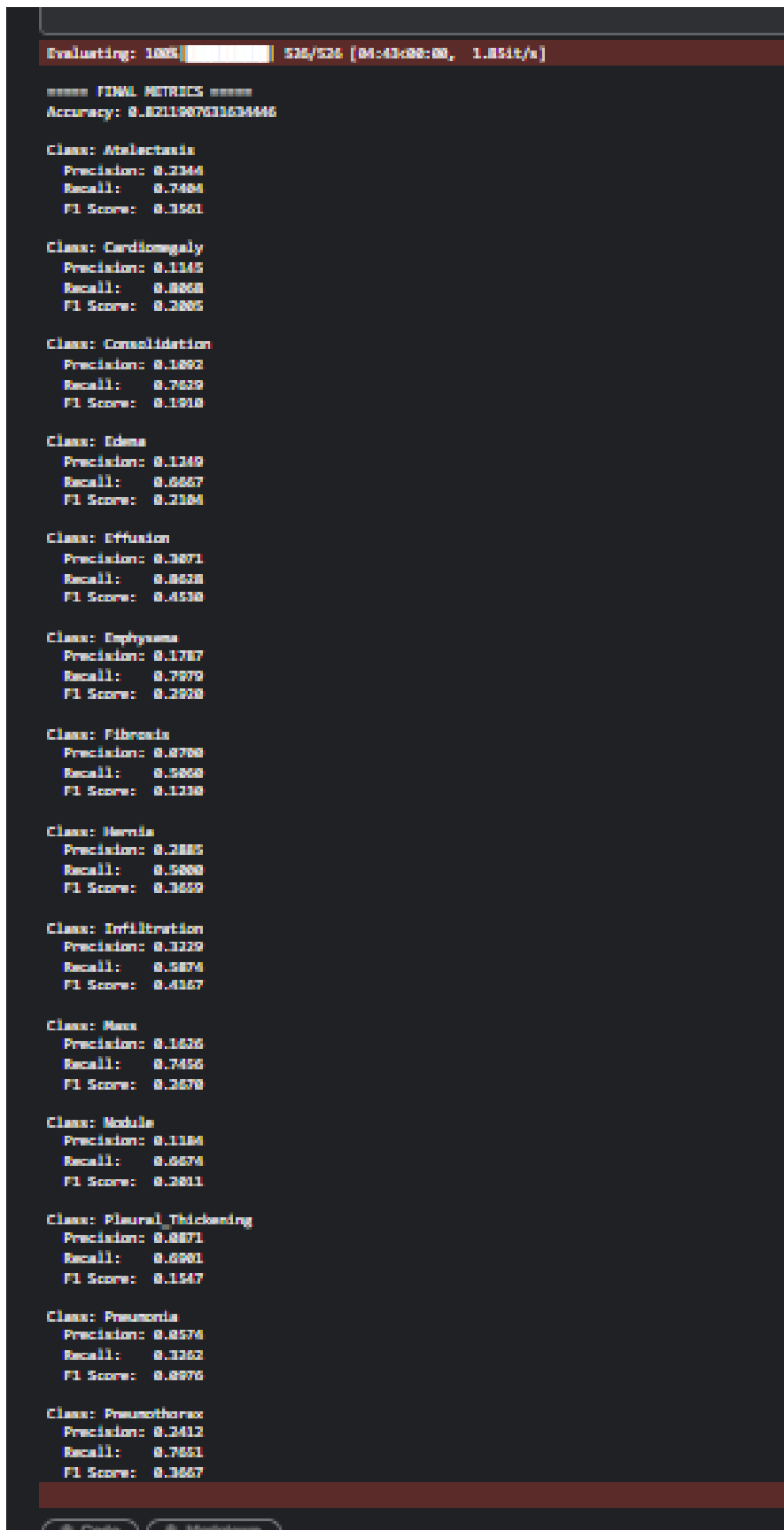


Figure 5: Per-class precision, recall, and F1-scores .

8. Explainability using Grad-CAM

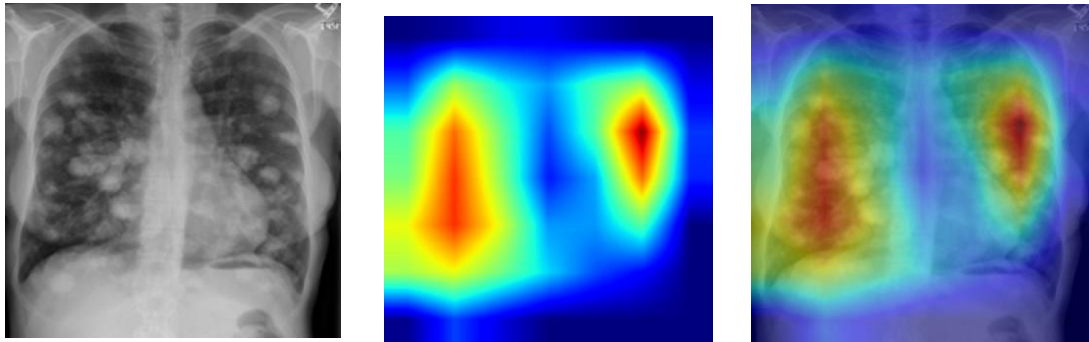


Figure 6: Grad-CAM visualization showing key activated regions.

9. Sample Prediction

```
print("\nBEST PREDICTION:")
print(f"{best_label} --> {confidence*100:.2f}%")

return best_label, confidence

# RUN
predict_single("/kaggle/input/sevedema/sev edema.png")

MODEL LOADED SUCCESSFULLY!

BEST PREDICTION:
Edema --> 78.59%

3]: ('Edema', 0.7858567833900452)
```

+ Code + Markdown

Figure 7: Sample model prediction (Edema detected with 78.59%).

10. Streamlit Web Application Deployment

An easy-to-use diagnostic interface was developed using Streamlit. The user uploads a chest X-ray and the app produces:

- Predicted disease and confidence
- Grad-CAM heatmap
- Overlay fused on original X-ray

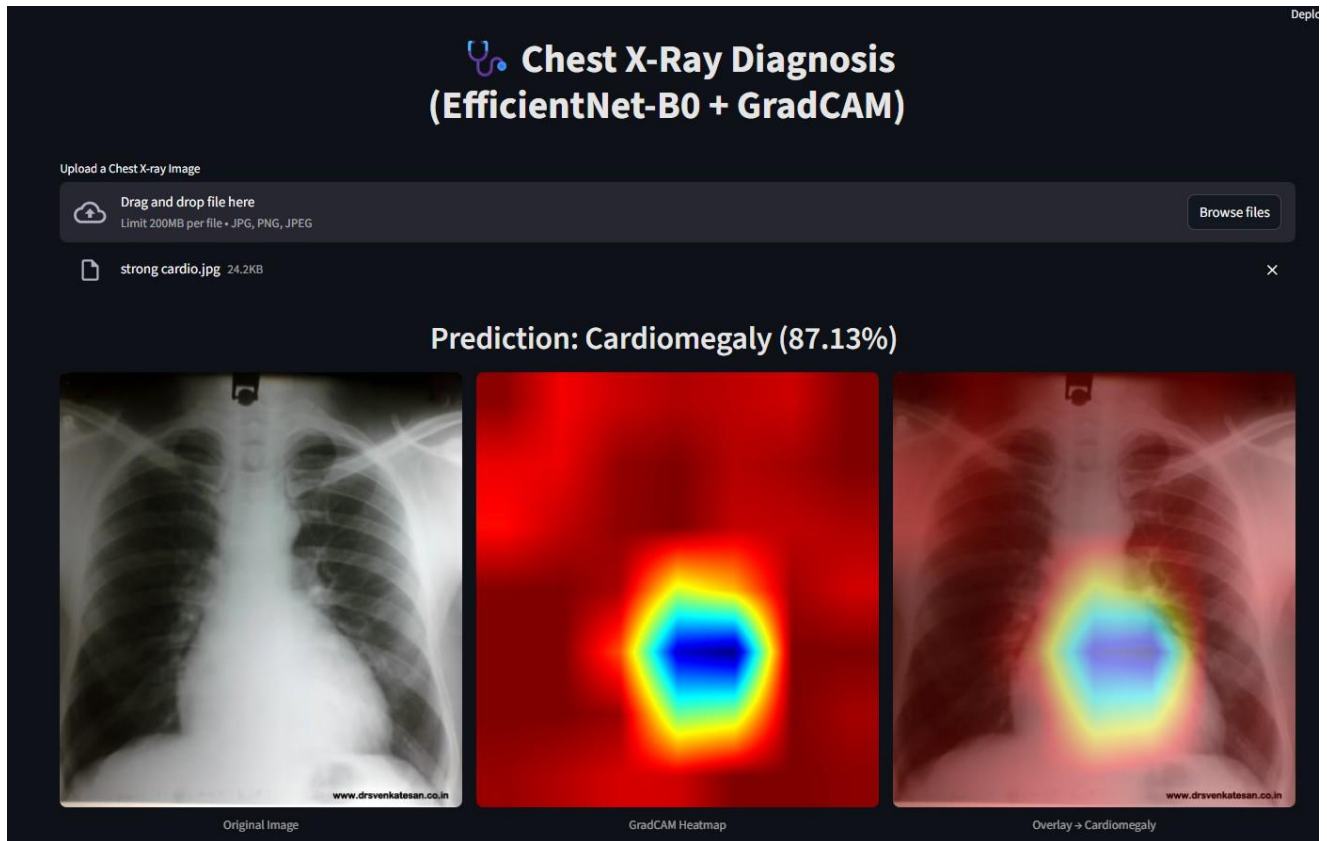


Figure 8: Streamlit app example — Cardiomegaly prediction (87.13%).

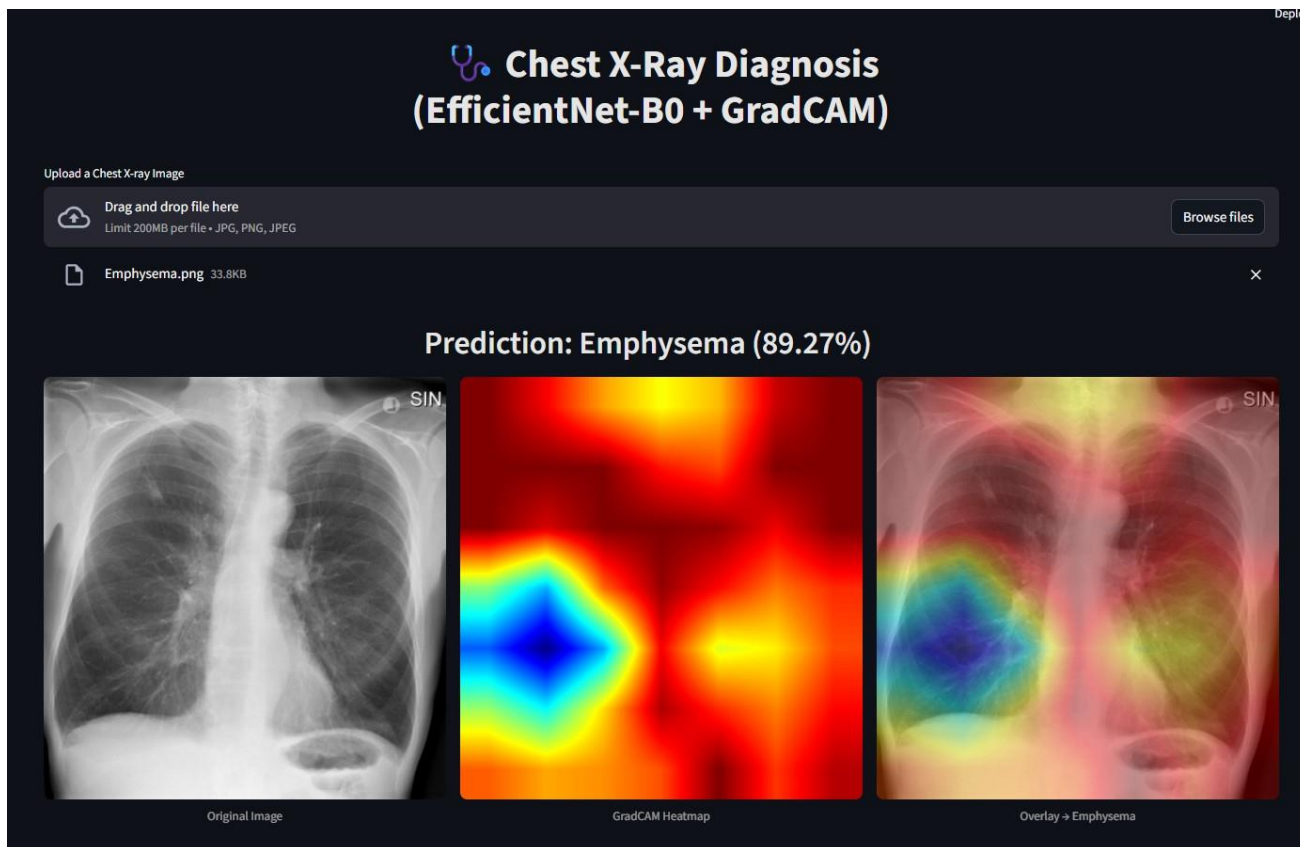


Figure 9: Streamlit app example — Emphysema prediction (89.27%).

11. Conclusion

The EfficientNet-B0-based multi-class classifier performs strongly across major disease categories and provides meaningful heatmap explanations. The Streamlit application makes the system usable for real-world demonstration and medical screening assistance.

12. Future Work

- Add lung segmentation preprocessing.
- Train larger EfficientNet variants (B3–B5).
- Use ensemble architectures for improved accuracy.
- Add model calibration techniques.
- Deploy the Streamlit app on Streamlit Cloud or ONNX Runtime.

References

- Tan, M. & Le, Q. (2019). EfficientNet.
- Selvaraju et al. (2017). Grad-CAM.
- NIH ChestXray14 Dataset.

- [PyTorch Documentation](#).
- [Albumentations Library](#).